VARIANCE-OPTIMAL HEDGING UNDER ROUGH VOLATILITY AND JUMPS

MAC F244: STOCHASTIC CALCULUS AND APPLICATIONS TO FINANCE
PROJECT REPORT (DRAFT)

1. Problem and summary.

Black–Scholes theory hedges an option perfectly by holding its delta, rebalanced continuously and at no cost. Real hedgers trade a few times a day at most, pay costs, and face markets that switch between calm and stress. We ask how an option should be hedged under these constraints: how often to rebalance, whether to trade more in stressed markets, and how much a better hedge ratio is worth. We answer this in two settings:

- *in simulation*, under a realistic model with rough volatility and clustered crash jumps [4], using the variance-optimal hedge computed by the published code of [2], which we reproduce;
- *on Indian market data*: NIFTY 50, India VIX and traded NIFTY options (2010–2026). Rules are calibrated on 2010–2018 and evaluated out of sample on 2019–2026.

2. Hedging: from complete to incomplete markets.

Throughout we work under the pricing measure $\mathbb{Q}$ with zero interest rate [1].

(a) Black–Scholes. Let $dS_t = \sigma S_t\, dW_t$ and let $C(t,S)$ be the call price, which solves $C_t + \frac{1}{2}\sigma^2 S^2 C_{SS} = 0$. By Itô's formula the dt-terms of dC cancel, so $dC = C_S\, dS$. Holding $\Delta = C_S$ shares therefore replicates the call exactly. This is the martingale representation theorem at work in a complete market.

(b) Discrete rebalancing. If Δ is reset only every h years, the hedge is frozen while the option's value moves nonlinearly. Let $\Gamma = C_{SS}$ and let R_k be the return over $[t_k, t_k + h]$. A second-order expansion with $C_t = -\frac{1}{2}\sigma^2 S^2 \Gamma$ gives the short-call hedging error over one interval:

$$\varepsilon_k \approx \tfrac{1}{2}\Gamma_k S_k^2\big(\sigma^2 h - R_k^2\big), \qquad \operatorname{Var}(\varepsilon_k) \approx \tfrac{1}{2}\Gamma_k^2 S_k^4 \sigma^4 h^2.$$

Summed over the T/h intervals, the extra variance is of order h: it roughly doubles when the interval doubles.

(c) Gamma P&L. If the delta is computed at an implied volatility σ_i while the stock moves with realised volatility σ_r, then $\mathbb{E}[R_k^2] \approx \sigma_r^2 h$. The delta-hedged seller therefore earns on average

$$\tfrac{1}{2}\,\Gamma S^2\big(\sigma_i^2 - \sigma_r^2\big)\, dt,$$

which is positive when implied volatility exceeds the volatility that follows.

(d) Incomplete markets and the minimum-variance hedge. With stochastic volatility or jumps there are more sources of risk than traded assets, so no stock position replicates the call. The best one can do is to minimise $\mathbb{E}^{\mathbb{Q}}[(Y - x - \int_0^T \vartheta_t\, dS_t)^2]$ for the payoff $Y = (S_T - K)^+$. Write $M_t = \mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t]$ (the option price) and decompose $Y = \mathbb{E}^{\mathbb{Q}}[Y] + \int_0^T \vartheta_t^*\, dS_t + L_T$, with the residual L uncorrelated with S (the Galtchouk–Kunita–Watanabe decomposition [5]). Then, for any (x, ϑ),

$$\mathbb{E}\Big[\Big(Y - x - \int_0^T \vartheta\, dS\Big)^2\Big] = (\mathbb{E}[Y] - x)^2 + \mathbb{E}\int_0^T (\vartheta^* - \vartheta)^2 d\langle S\rangle + \mathbb{E}[L_T^2],$$

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so the optimum is $x = \mathbb{E}^{\mathbb{Q}}[Y]$ (the price) and

$$\vartheta_t^* = \frac{d\langle M, S\rangle_t}{d\langle S\rangle_t},$$

a regression of option moves on stock moves. $\mathbb{E}[L_T^2]$ is the risk no stock position can remove. *Example (Heston):* for $dS = \sqrt{v}\, S\, dW$ and $dv = \kappa_v(\theta - v)\, dt + \xi\sqrt{v}\, dB$ with $d\langle W, B\rangle = \rho\, dt$, write $M_t = C(t, S_t, v_t)$. Two-variable Itô gives $d\langle M, S\rangle = (C_S\, vS^2 + C_v\, \rho\xi vS)\, dt$ and $d\langle S\rangle = vS^2\, dt$, so

$$\vartheta^* = C_S + \frac{\rho\, \xi\, C_v}{S}.$$

With $\rho < 0$ (volatility rises when prices fall) and $C_v > 0$, the variance-optimal hedge holds fewer shares than the delta.

(e) The simulation model. For simulation we use the rough Hawkes–Heston model [4]. It is a Heston-type model whose variance has long memory (it depends on all past shocks through a power-law kernel, giving "rough" paths) and whose price has crash jumps that become more likely after a jump. Wang and Wang [2] derive ϑ^* for this model. It has the covariance structure of (d) plus a jump term, and it is computed with Fourier and Riccati–Volterra methods. We use the authors' published implementation [3].

(f) Discrete trading with costs. We trade a target hedge (ϑ^* or a delta) only at dates $t_0 < \cdots < t_{N-1}$, holding φ_k shares on $(t_k, t_{k+1}]$. With a proportional cost κ per unit of traded notional, the short-call P&L is

$$e = x + \sum_{k=0}^{N-1} \varphi_k(S_{k+1} - S_k) - \kappa \sum_{k=0}^{N-1} S_k|\varphi_k - \varphi_{k-1}| - \kappa S_N|\varphi_{N-1}| - Y, \qquad \varphi_{-1} = 0.$$

A rebalancing rule chooses when φ is reset to its target. The rules are: *fixed* (every m steps); *regime-aware* (every m_{stress} steps in stress, m_{calm} otherwise); and a *no-trade band* (trade when the target moves more than h shares). Every rule uses only information up to t_k.

3. Replication.

Running the official code [3] (50,000 paths, $T = 0.25$, ATM call) reproduces its results (Table 1). Measured as the reduction in hedging-error variance against not hedging, it gives 0.828 for the variance-optimal (GKW) hedge and 0.732 for the Black–Scholes delta. These numbers are official-code output rather than results printed in [2]. The small differences from the stored values come from a different random stream and lie within one standard error. The paper's own convergence tables also reproduce.

Hedge	Official output	Our rerun
Variance-optimal (GKW) hedge	0.824	0.828
Black–Scholes delta (implied vol)	0.726	0.732

TABLE 1. Variance reduction vs. no hedge, 50,000 paths, 512 steps.

4. Discrete rebalancing with costs (simulation).

We reuse the official market simulation and hedge, and change only the trading dates and the costs. All rules are evaluated on the same paths, so comparisons use paired standard errors. With fewer rebalancing dates the variance reduction falls for both hedges, as Section 2(b) predicts. The GKW advantage over the BS delta *grows* (Table 2): the better hedge ratio matters more when the position is held longer. With costs, GKW beats BS at every cost level (z between -5.1 and -6.2). To compare rules fairly, each band or regime rule is set against a randomised mixture of the two neighbouring fixed rules with the same expected turnover, and hence the same expected cost. At low turnover these rules reduce the mean squared error by 6%–31% at $\kappa = 10$ bp, and the sign agrees on an independent seed in 15/16 cases. At the cost-optimal point, however, frequent trading is optimal and the rules nearly coincide. Against the

best fixed rule, the best smart rule changes the mean squared error by only -1.2% ($z = -0.5$) for GKW and -5.9% ($z = -1.7$) for BS.

Rebalancing dates	GKW hedge	BS delta
512	0.828 (0.012)	0.732 (0.026)
64	0.806 (0.013)	0.670 (0.031)
16	0.738 (0.016)	0.541 (0.033)
4	0.583 (0.021)	0.333 (0.036)
1	0.374 (0.019)	0.108 (0.036)

TABLE 2. Variance reduction (standard error) against the number of rebalancing dates, $\kappa = 0$.

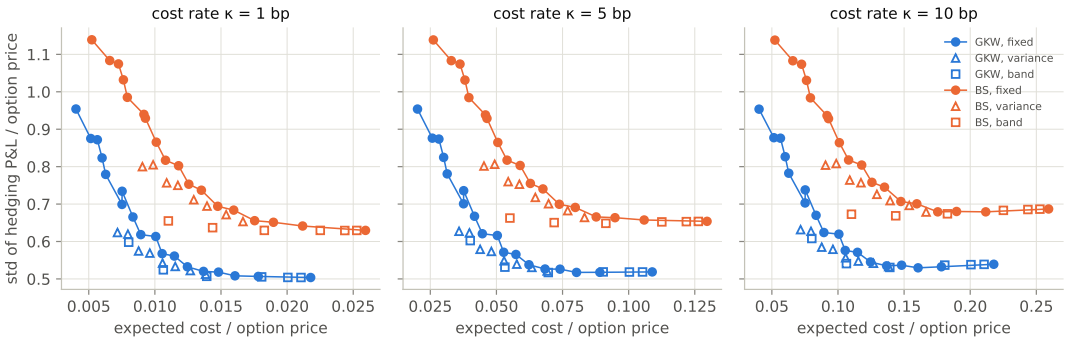


FIGURE 1. Simulation: standard deviation of the P&L against expected cost (both relative to the option price) for each rule.

5. Data and regimes.

We use the official NIFTY 50 closes (NSE Indices, 4,740 days; Yahoo Finance agrees on common days but misses 47) and India VIX from Yahoo, patched and verified against NSE (4,032 days from July 2010, all 196 months checked) [9, 10]. Regimes are estimated on 2010–2018 only. The main label is India VIX above 20.16, with hysteresis down to 17.88 so that the label does not flicker. The second label is 21-day realised volatility above 17.4%. Each label at t uses data up to t only, which we check by truncating the data at t. Out of sample, stress labels are followed by 1.7–2.0 times higher realised volatility (Table 3, Figure 2). A third, statistical label (a two-state Markov-switching model of returns [7]) is used only as a robustness check and gives the same picture (ratio 1.74, lower 95% bound 1.40).

Variant	VIX (hysteresis)	21-day realised vol
Main (train to 2018)	1.94 [1.58, 2.44]	1.79 [1.37, 2.41]
Train to 2017	1.94 [1.57, 2.46]	1.80 [1.37, 2.38]
Train to 2019	2.01 [1.65, 2.54]	1.85 [1.40, 2.47]
Start 2008 (unverified VIX)	2.25 [1.52, 3.35]	1.86 [1.32, 2.64]

TABLE 3. Test period 2019–2026: realised volatility over the next 5 days after a stress label, divided by that after a calm label [block-bootstrap 95% CI].

6. Backtest on NIFTY with synthetic options.

At each close we sell a 21-day ATM call priced at India VIX (Black–Scholes [6]) and hedge with the delta at the current VIX, giving 1,914 test options with a mean premium of 197 bp.

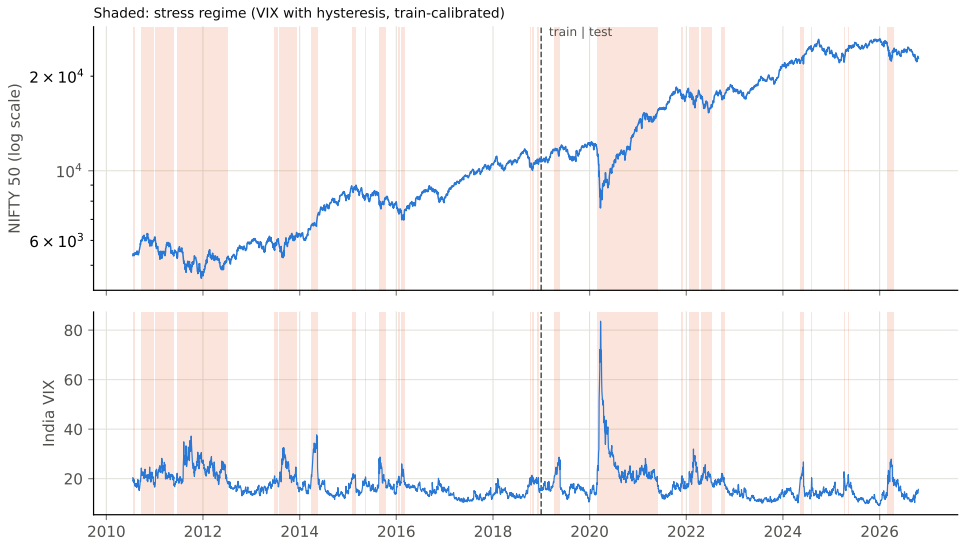


FIGURE 2. NIFTY 50 and India VIX; shaded: stress (VIX with hysteresis).

The seller's cost-free mean P&L is about +30 bp. This is the gamma P&L of Section 2(c), since implied volatility usually exceeded the volatility that followed. India VIX was above the next month's realised volatility on 79% of test days (mean 17.1% against 14.7%). In variance terms, implied exceeded realised by a factor 1.36 outside the COVID window, while the crash reversed it (factor 0.73). Because this mean reflects the volatility premium rather than hedging quality, we rank rules by $J = \mathrm{Var}(e) + (\mathbb{E}[\text{cost}])^2$, which equals the simulation criterion when the premium is fair. Options overlap, so standard errors use a moving-block bootstrap [8]. Daily rebalancing is the best rule on the training period at every cost up to 20 bp, and the regime rule chosen on train is *worse* on test (Table 4). At equal turnover, regime-aware rules appear better on the full test period, but this comes from the 2020 COVID crash: excluding it, or using 63-day options, the advantage disappears. Rules driven by inverted labels are worse than those driven by the real labels, so trading less in stress clearly hurts. Trading more in stress, however, does not reliably beat a fixed schedule.

	Picked on train vs daily		Equal turnover, 5 bp		
Variant	5 bp	20 bp	beat fixed	med. z	beat placebo
21-day, full test	+19.5% (2.9)	+7.9% (1.6)	42/48	−1.1	46/48
21-day, excl. COVID window	+30.0% (2.7)	+9.8% (1.5)	20/48	0.2	34/48
63-day options	+8.1% (3.3)	+2.7% (1.0)	19/48	0.3	35/48

TABLE 4. Synthetic-option backtest, test period. Left: change in J of the regime rule picked on train vs. daily rebalancing (z). Right: regime rules (4 labels × 12 settings) vs. fixed rules at equal turnover, and how often they beat the inverted-label placebo.

7. Backtest with traded NIFTY options.
We repeat Section 6 with traded prices from the NSE F&O bhavcopies [11] (4,031 trading days, 2010–2026). The data pass basic checks:

- at all 194 monthly expiries the expiring future settles exactly at the NIFTY close;
- near-ATM put–call parity holds within ±28 bp (1st–99th percentile, 136,118 pairs);
- only 0.28% of traded near-ATM calls violate no-arbitrage bounds;

- ATM implied volatility has correlation 0.96 with India VIX.

Expiries moved for holidays are detected from the data. At each close we sell the monthly call closest to 21 trading days from expiry, at the strike nearest the futures price and at its traded close. We hedge with the same-expiry future at the Black–76 delta [6], using the option's own implied volatility; both contracts settle on the index at expiry. This gives 1,898 test options (mean premium 179 bp). Daily hedging reaches a J-RMSE of 57.7 bp, against 64.0 bp with synthetic options, and the cost-free mean P&L is +18 bp. Again this is gamma P&L: the option's implied volatility exceeded NIFTY's realised volatility over its life for 73% of options (mean 15.7% against 14.5%; variance factor 1.20 outside the COVID window, 0.68 inside it). Using India VIX instead of the option's own implied volatility in the delta changes J by only -4.3% ($z = -1.5$). The conclusions of Section 6 carry over (Table 5, Figure 3). Daily rebalancing is best on the training period, and the regime rule picked on train is worse on test. The equal-turnover advantage of regime-aware rules holds on the full test period but not once the COVID window is excluded. Real labels again beat the inverted placebo.

	Picked on train vs daily		Equal turnover, 5 bp		
Variant	5 bp	20 bp	beat fixed	med. z	beat placebo
Own implied vol	+35.0% (2.1)	+15.1% (1.4)	48/48	−1.0	48/48
Own IV, excl. COVID window	+51.8% (2.0)	+18.8% (1.3)	17/48	0.4	41/48
India VIX delta	+33.7% (2.1)	+16.3% (1.5)	47/48	−1.0	46/48

TABLE 5. Traded-option backtest, test period (same layout as Table 4).

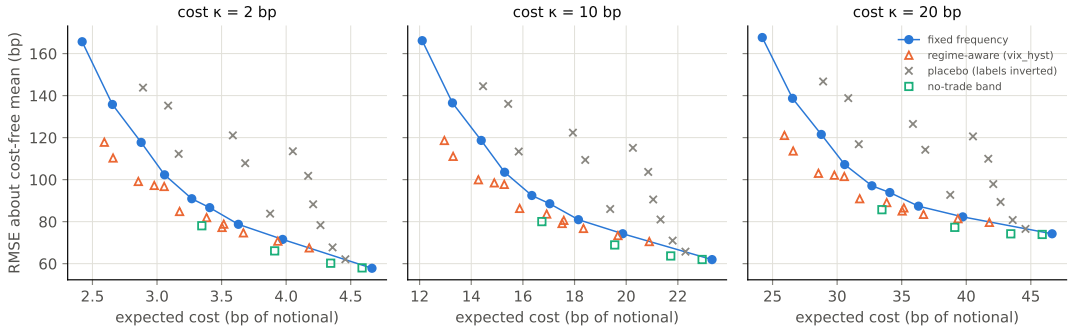


FIGURE 3. Traded NIFTY options, test period 2019–2026 (COVID included): RMSE about the cost-free mean against expected cost, for each rule.

8. Conclusions and limitations.

Hedge ratio. Trading the variance-optimal hedge discretely keeps most of its advantage over the Black–Scholes delta, and the advantage grows as rebalancing becomes coarser. *Timing.* In simulation and on NIFTY data, regime-aware rebalancing helps only when turnover must stay low. At realistic NIFTY costs (up to 20 bp) daily rebalancing is optimal, and the regime rule chosen in sample does worse out of sample. The real-data gains at equal turnover come from a single episode (COVID 2020), so they should not be read as a general effect. *Profit and loss.* The delta-hedged seller's average gain on NIFTY is the gamma P&L of Section 2(c): implied volatility usually exceeded realised volatility, except in the COVID crash.

Limitations.

- The simulation uses one calibration of the model, under $\mathbb{Q}$.
- The variance-optimal hedge is computed with the authors' published code.

- The model is not calibrated to NIFTY, so the real-data hedges are Black–Scholes and Black–76 deltas.
- Prices are daily closes, which are not perfectly synchronous.
- Costs are proportional and do not depend on the regime, and margin financing is ignored.
- The test period contains essentially one crisis.

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Group members.

No.	Name
1	Dhyan Laad
2	Natasha Nalwaya
3	Shrirang Bhale
4	Tanmai Chekuri
5	Shivam Mahto